

SOLUTION OF MTH643 BY MOIN AKHTAR

Calculate the area and circumference of a circle with a radius of 4 cm.

SOLUTION:

```
>> syms r;
```

```
>> r=4;
```

```
>> a=pi*r^2
```

a =

50.2655

```
>> c=a/2
```

c =

25.1327

① Find the value of the following polynomials at $x = 2$ using **subs** command:

- $p(x, y) = x^2 + 3y^2x + 10$

- $u(x) = x^3 + 4x + 10$

SOLUTION1:

```
syms x y
```

```
p(x,y)=(x^2+3*x*y^2+10)
```

```
p(x, y) =
```

```
x^2 + 3*x*y^2 + 10
```

```
subs(p(x,y),x,2)
```

```
ans =
```

```
6*y^2 + 14
```

Solution2:

```
syms x
```

```
u(x)=x^3+4*x+10
```

```
u(x) =
```

```
x^3 + 4*x + 10
```

```
subs(u(x),x,2)
```

```
ans =
```

```
26
```

② Find the value of the following trigonometric functions at $x = \frac{\pi}{2}$ using sym command:

- $\cos(x)$, $\sin(x)$, $\tan(x)$.

Solution:

```
x=sym(pi/2)
```

```
x =
```

```
pi/2
```

```
cos(x)
```

```
ans =0
```

```
sin(x)
```

```
ans =1
```

```
tan(x)
```

```
ans =Inf
```

① Expand the following algebraic expressions using **expand** command:

- $(x + y)^{10}$
- $(x + y)^3 + (x + y)^2$

② Simplify the following quantities:

- $2 + 5 - 4 + 10^{10} + 7^2$.
- $2/5 + 4/10 - 100^2$

SOLUTION:

Part 1

```
close all
```

```
clear all
```

```
syms x y
```

```
expand((x + y)^10)
```

```
ans =
```

```
x^10 + 10*x^9*y + 45*x^8*y^2 + 120*x^7*y^3 + 210*x^6*y^4 + 252*x^5*y^5 +  
210*x^4*y^6 + 120*x^3*y^7 + 45*x^2*y^8 + 10*x*y^9 + y^10.
```

Part 2

syms x y

expand((x + y)^3+(x + y)^2)

ans =

$x^3 + 3x^2y + x^2 + 3xy^2 + 2xy + y^3 + y^2.$

Part 3

$2+5-4+10^{10}+7^2$

ans =1.0000e+10

Part 4

$2/5+4/10-100^2$

ans = -9.9992e+03

Solve the following equations using MATLAB **solve** command

• $x^2 + 2x - 4 = 0.$

• $x^3 + 5x^2 + 4x + 3 = 0$

SOLUTION:

```
solve('x^2 + 2*x - 4 = 0')
```

ans =

$5^{1/2} - 1$
 $- 5^{1/2} - 1$

```
solve('x^3 + 5*x^2 + 4*x + 3 = 0')
```

ans =

$- 13/(9*(151/54 - 173^{1/2}/6)^{1/3}) - (151/54 - 173^{1/2}/6)^{1/3} - 5/3$
 $13/(18*(151/54 - 173^{1/2}/6)^{1/3}) + (151/54 - 173^{1/2}/6)^{1/3}/2 + (3^{1/2})*(13/(9*(151/54$
 $- 173^{1/2}/6)^{1/3}) - (151/54 - 173^{1/2}/6)^{1/3})*i)/2 - 5/3$
 $13/(18*(151/54 - 173^{1/2}/6)^{1/3}) + (151/54 - 173^{1/2}/6)^{1/3}/2 - (3^{1/2})*(13/(9*(151/54$
 $- 173^{1/2}/6)^{1/3}) - (151/54 - 173^{1/2}/6)^{1/3})*i)/2 - 5/3$

Solve the following equations for x using **solve** command

- $x + \log(y) = 3.$
- $x + 10y = 6.$

SOLUTION:

```
solve('x+log(y)=3')
```

```
ans =
```

```
3 - log(y)
```

```
>> solve('x+10*y=6')
```

```
ans =
```

```
6 - 10*y
```

Find the solution of the following equation using **fzero** command near to $x = 3$

- $\exp(-x) = \sin(x).$

SOLUTION:

```
f=@(x)exp(-x)-sin(x);  
fzero(f,3)
```

```
ans =
```

```
3.0964
```

Solve the following system of equations using **solve** command

- $x + y^2 = 2, y - 3x = 7.$

- $x + y^2 = 2, 2y^2 - 3x = 7.$

SOLUTION:

1.

`[x,y]=solve('x+y^2=2','y-3*x=7')`

x =

$$\begin{aligned} &157^{(1/2)}/18 - 43/18 \\ &- 157^{(1/2)}/18 - 43/18 \end{aligned}$$

y =

$$\begin{aligned} &157^{(1/2)}/6 - 1/6 \\ &- 157^{(1/2)}/6 - 1/6 \end{aligned}$$

2.

`[x,y]=solve('x+y^2=2','2*y^2-3*x=7')`

x =

$$\begin{aligned} &-3/5 \\ &-3/5 \end{aligned}$$

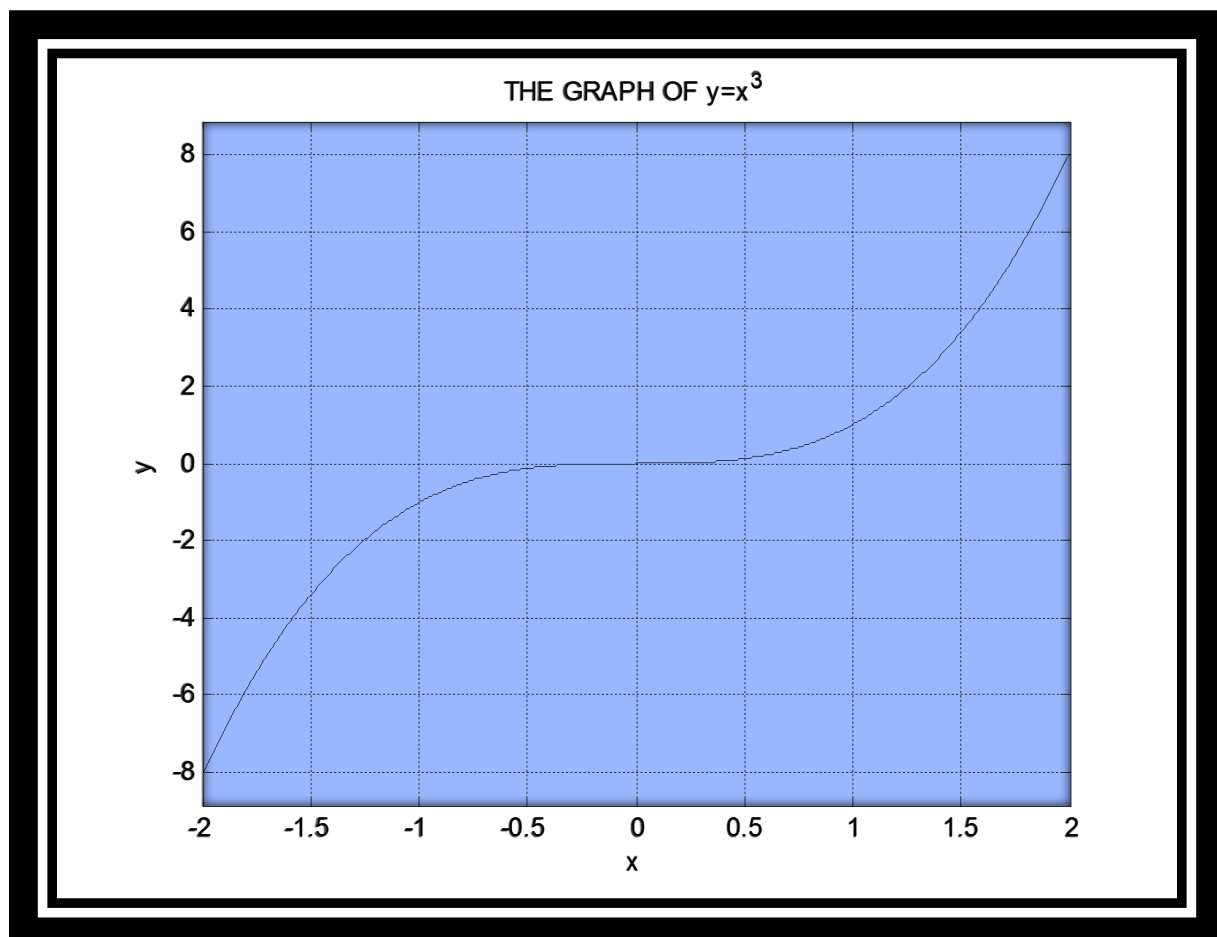
y =

$$\begin{aligned} &65^{(1/2)}/5 \\ &-65^{(1/2)}/5 \end{aligned}$$

- The graph of function x^3 can also be plotted by using an anonymous function as the argument of **ezplot**. For this case the input will be:

SOLUTION:

```
>> syms x y;  
>> y=ezplot(x^3,[-2,2]);  
>> xlabel('x');  
>> ylabel('y');  
>> title('THE GRAPH OF y=x^3');  
>> grid
```

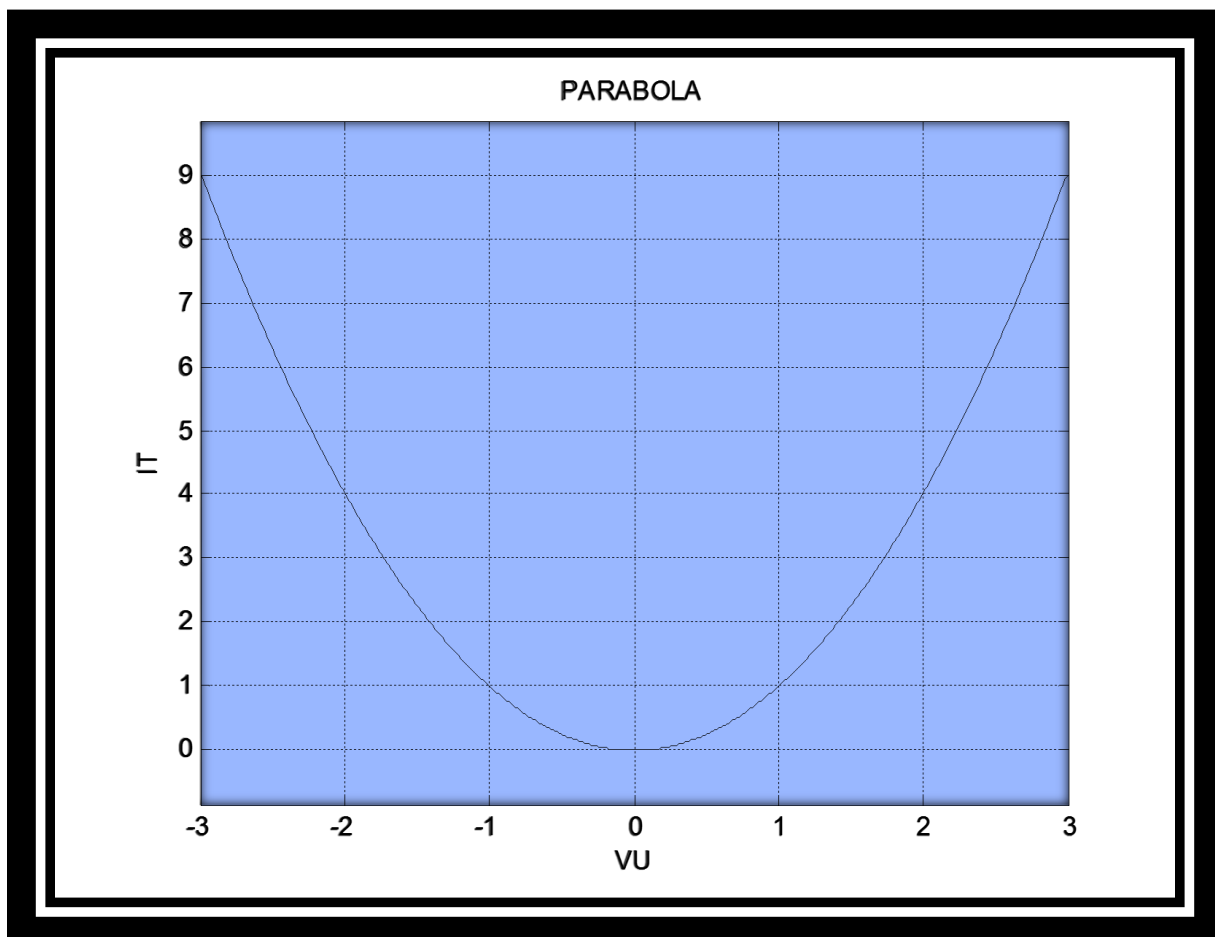


Example:

- Plot the graph of the function $y = x^2$ on $[-3, 3]$ using ezplot command considering x and y to be symbolic variables.
- Label the horizontal axis as "VU" and vertical axis as "IT".
- Keep the title of the graph as "Parabola"

SOLUTION:

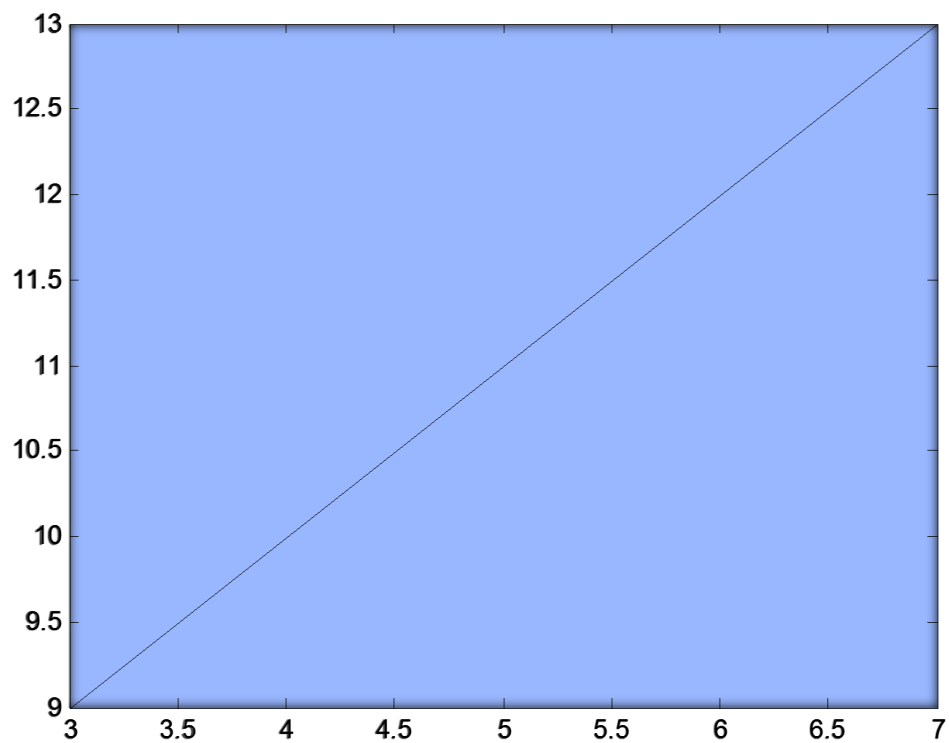
```
>> syms x y;  
>> y=ezplot(x^2,[-3,3]);  
>> xlabel('VU');  
>> ylabel('IT');  
>> title('PARABOLA');  
>> grid
```



- Under **Plot** command the vectors $X = [3 \ 5 \ 7]$ and $Y = [9 \ 11 \ 13]$ are connected by line segments as (3,9) to (5,11) to (7,13).

SOLUTION:

```
>> X = [3 5 7];  
>> Y = [9 11 13];  
>> plot(X,Y)
```

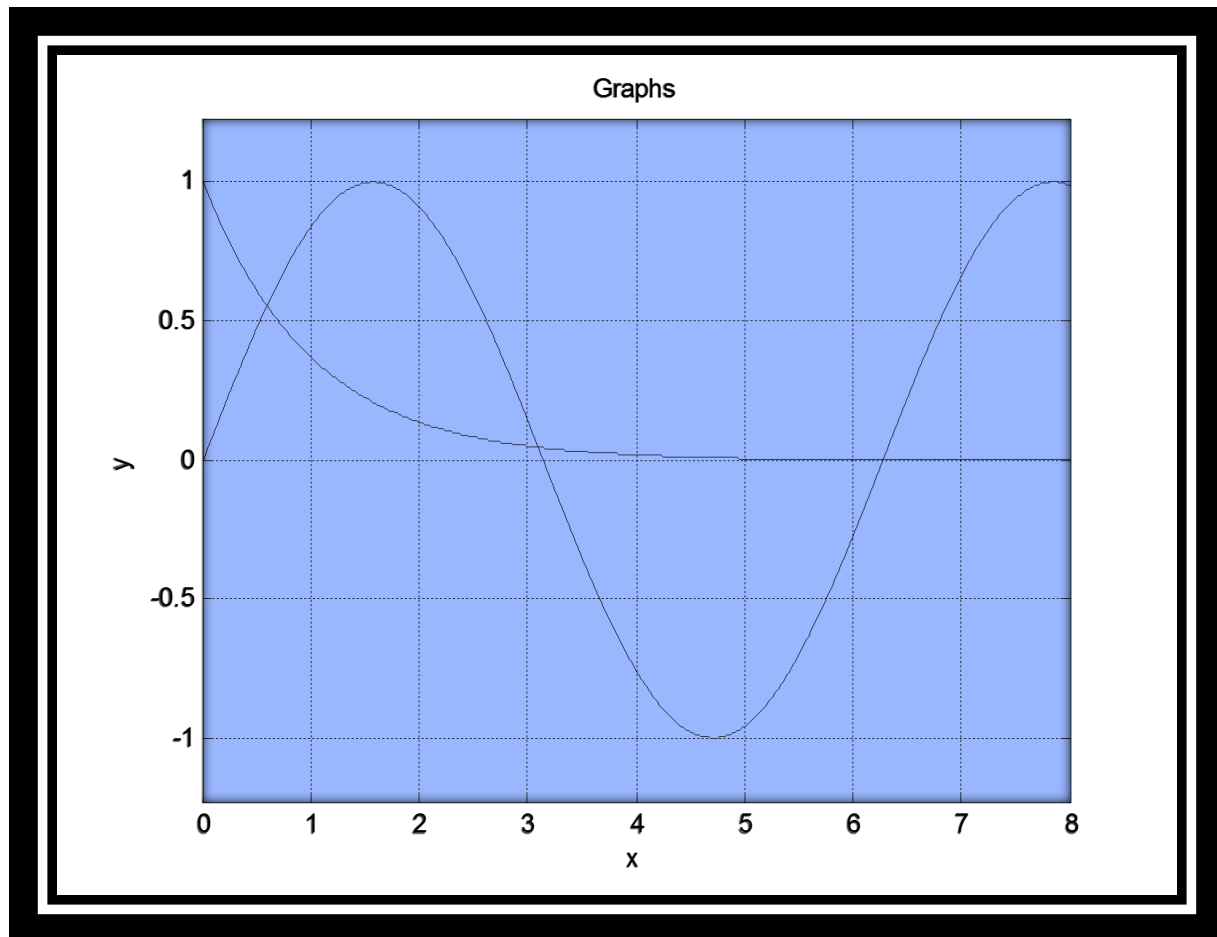


Graphs of Multiple Curves

- Each time we execute a plotting command, MATLAB erases the old plot and draws a new one.
- But if we want to see the two or more plots, then use command **hold on**.
- This command instructs MATLAB to retain the old graphics and draw any new graphics on top of the old.
- This command remains in effect until we type **hold off**.

SOLUTION:

```
>> syms x;  
>> ezplot(exp(-x),[0,8]);  
>> hold on  
>> ezplot(sin(x),[0,8]);  
>> hold off  
>> title('Graphs')  
>> xlabel('x');  
>> ylabel('y');  
>> grid;
```



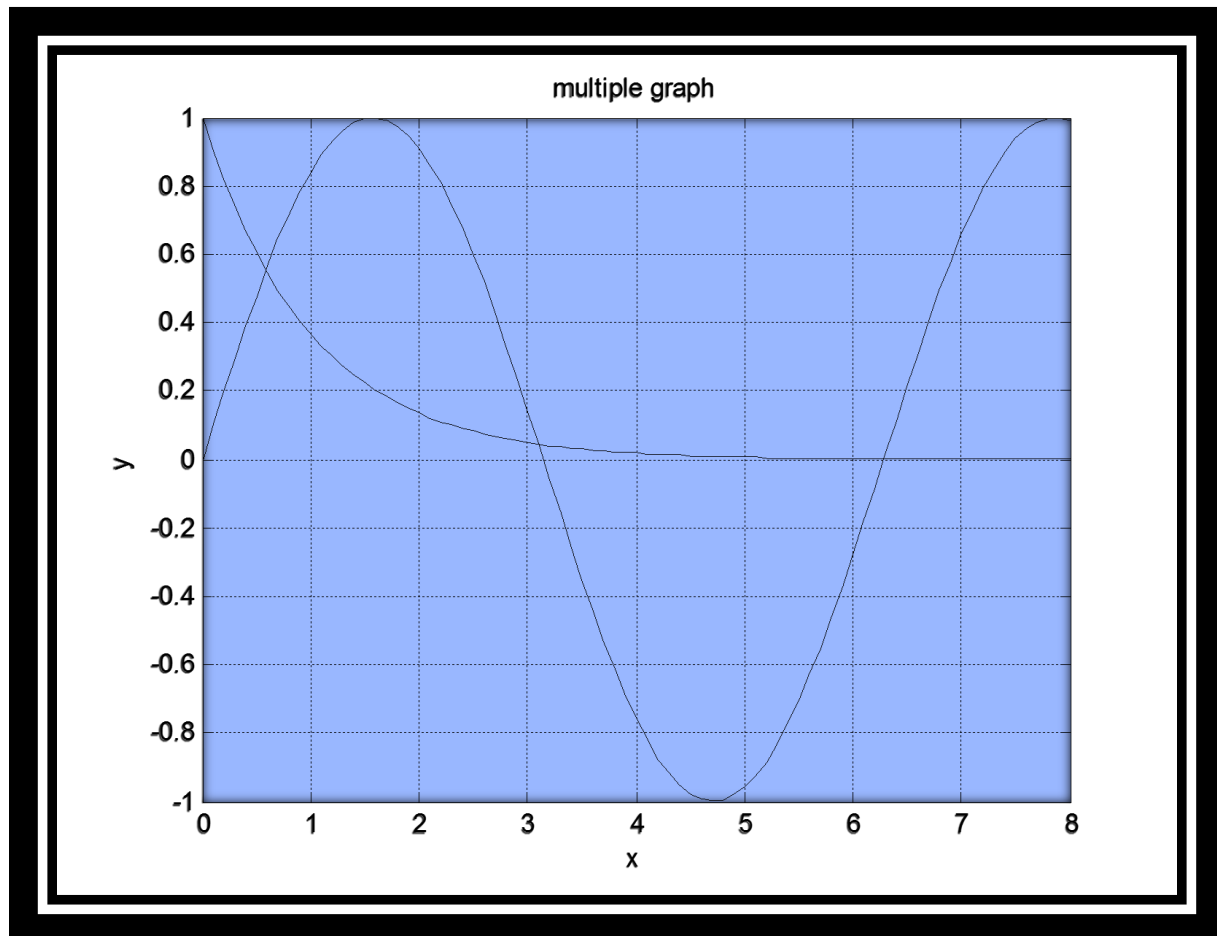
Graphs of Multiple Curves

Example:

- Plot the graphs of e^{-x} and $\sin(x)$ on the interval $[0, 8]$ using the **plot**, **hold on**, and **hold off** commands.
- Label the horizontal axis with x and vertical axis with y .
- Finally, gives the title "Multiple Graphs".

SOLUTION:

```
>> X=0:0.1:8;  
>> plot(X,exp(-X))  
>> hold on  
>> plot(X,sin(X))  
>> hold off  
>> title('multiple graph')  
>> xlabel('x')  
>> ylabel('y')  
>> grid
```



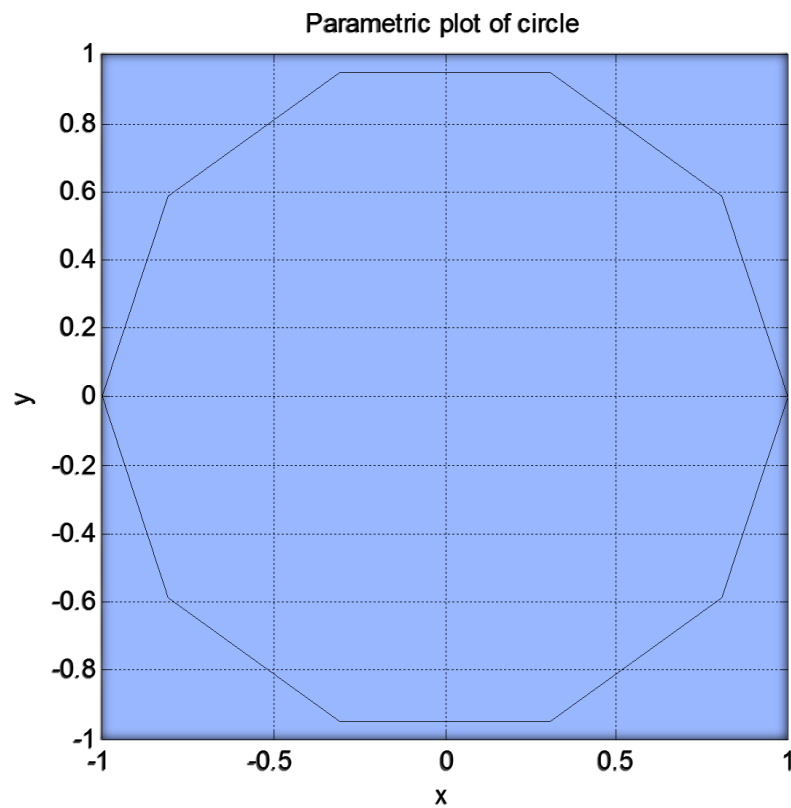
Parametric Plots

Example:

- Plot the parametric graph of the circle with center at $(0,0)$, radius 1, and with increment of 0.01 in the domain set.
- Label the horizontal axis with x and vertical axis with y .
- Make the shape of the graph square.
- Keeps the title of the graph "Parametric plot of the circle".

SOLUTION:

```
>> t=0:0.1:1;  
>> plot(cos(2*pi*t),sin(2*pi*t))  
>> xlabel('x');  
>> ylabel('y');  
>> title('Parametric plot of circle');  
>> grid;  
>> axis square;
```

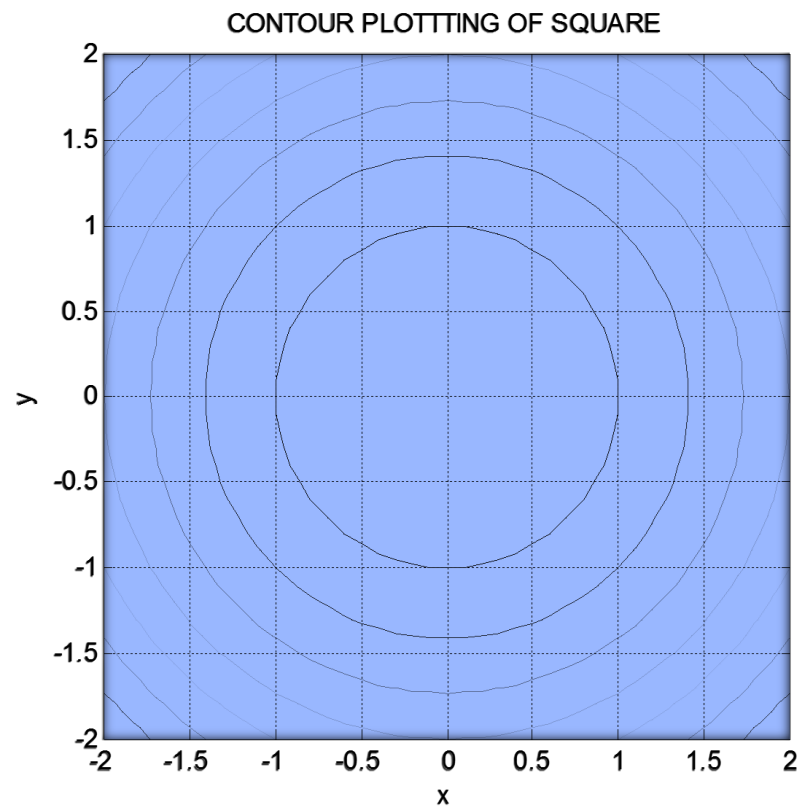


Example:

- Plot the contour plots of the expression $x^2 + y^2$ using **meshgrid** and **contour** commands.
- Label the horizontal axis with x and vertical axis with y .
- Make the shape of the graph square.
- Keeps the title of the graph "Contour plotting of squares".

SOLUTION:

```
>> [X,Y]=meshgrid(-2:0.1:2,-2:0.1:2);  
>> contour(X,Y,X.^2+Y.^2);  
>> axis square;  
>> xlabel('x');  
>> ylabel('y');  
>> grid;  
>> title('CONTOUR PLOTTING OF SQUARE');
```

- Plot the graph of the function $x^3 + x^2 + 4$ on the interval $[-2, 2]$, using **ezplot** command.

Label the horizontal axis with x and vertical axis with y .
Make the shape of the graph square.

Keeps the title of the graph "The Graph of $x^3 + x^2 + 4$ ".

Keeps the layout of the graph as grid.

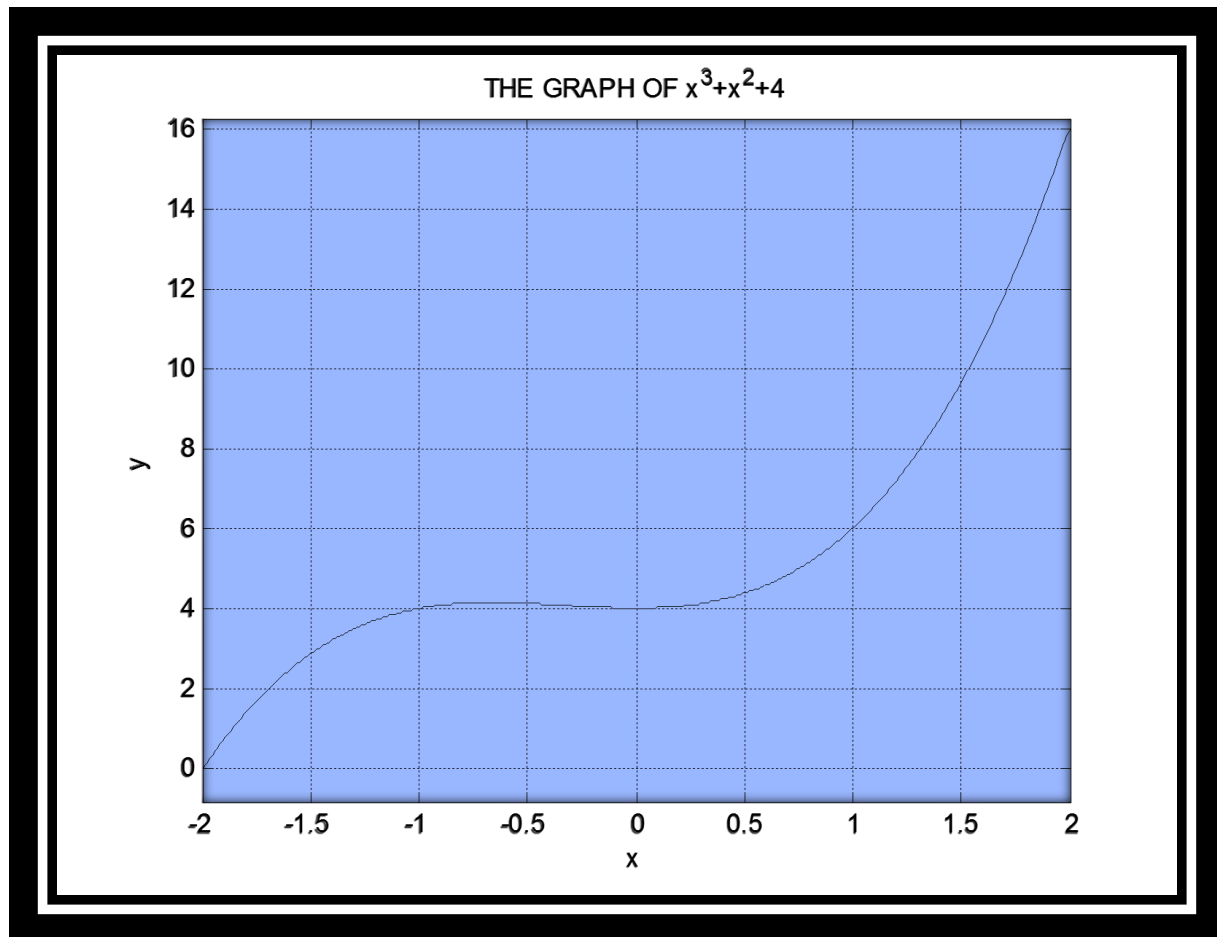
SOLUTION:

METHOD.1:

```
>> ezplot(@(x)(x^3+x^2+4),[-2,2])
>> xlabel('x');
>> ylabel('y');
>> title('THE GRAPH OF x^3+x^2+4');
>> grid;
```

METHOD.2:

```
>> syms x;
>> p=ezplot(x^3+x^2+4,[-2,2]);
>> xlabel('x');
>> ylabel('y');
>> title('THE GRAPH OF x^3+x^2+4');
>> grid;
```



- Plot the graph of the function $\sin(x) + \cos(x)$ on the interval $[0, 1]$, using **plot** command.

Label the horizontal axis with x and vertical axis with y .

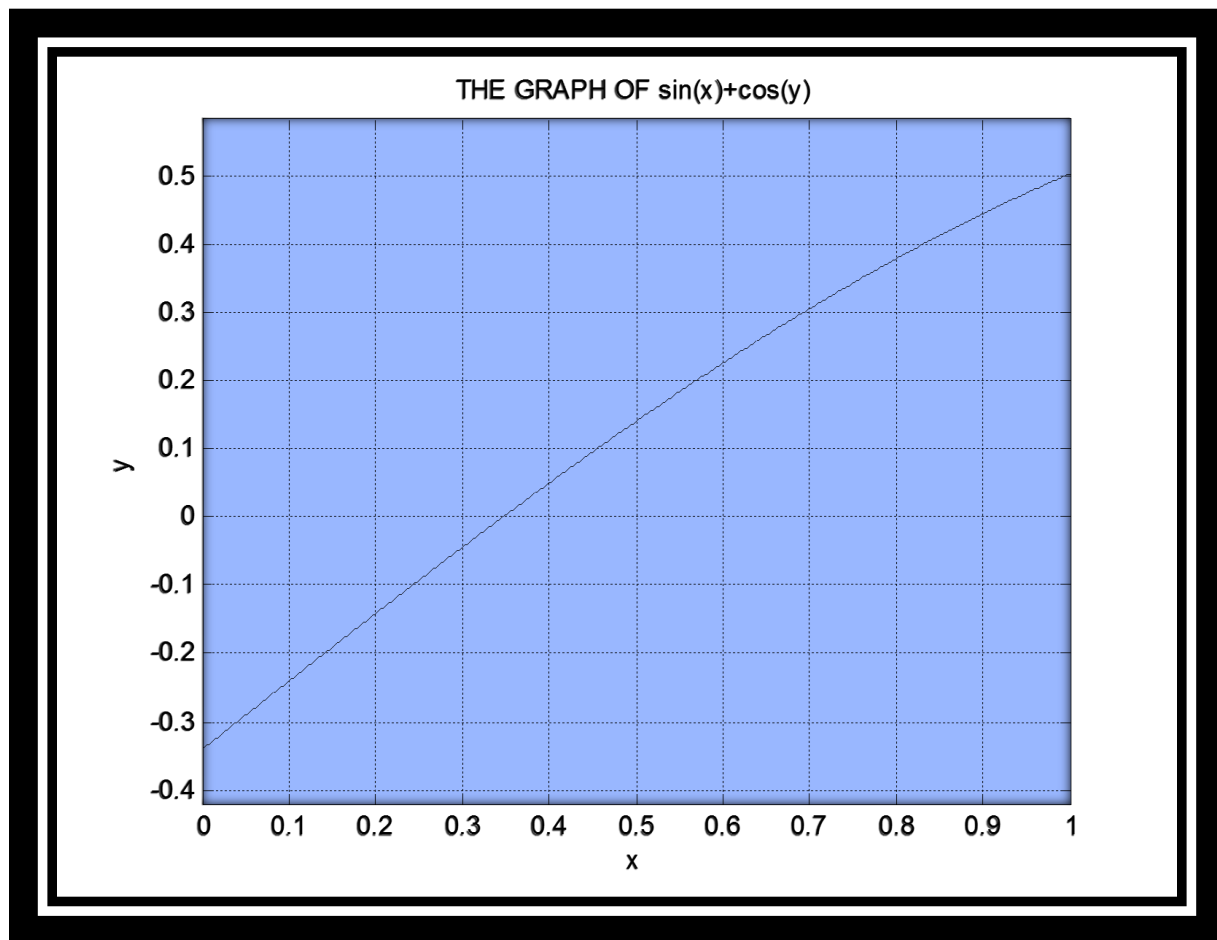
Make the shape of the graph square.

Keeps the title of the graph "The Graph of $\sin(x) + \cos(x)$ ".

Keeps the layout of the graph as grid.

SOLUTION:

```
>> syms x;  
>> y=ezplot(sin(x)+cos(y),[0,1]);  
>> xlabel('x');  
>> ylabel('y');  
>> title("THE GRAPH OF sin(x)+cos(y)");  
>> grid;  
—
```



- Plot the multiple curves of the functions $\sin(x) + \cos(x)$ and $e^{-x} + x^2$ on the interval $[0, 4]$, using **plot** command and without the use of **hold on** and **hold off** commands.

Label the horizontal axis with x and vertical axis with y .
Make the shape of the graph square.

Keeps the title of the graph "The Graph of Multiple curves using plot command".

Keeps the layout of the graph as grid.

SOLUTION:

```
>> syms x e;
>> ezplot(e^-x+x^2,[0,4]);
>> hold on;
>> ezplot(sin(x)+cos(x),[0,4]);
>> hold off;
>> xlabel('x');
>> ylabel('y');
>> title('THE GRAPH OF MULTIPLE CURVE');
>> grid;
```

